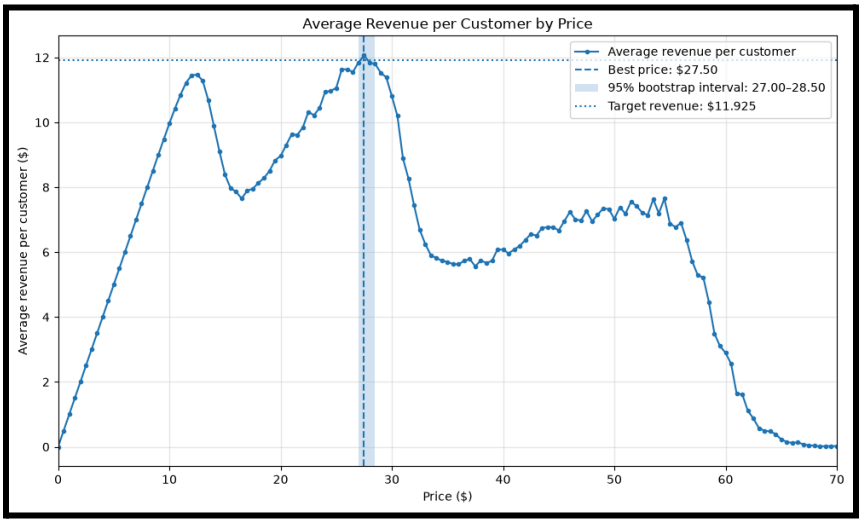
Optimal Price Report
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1. Problem Summary

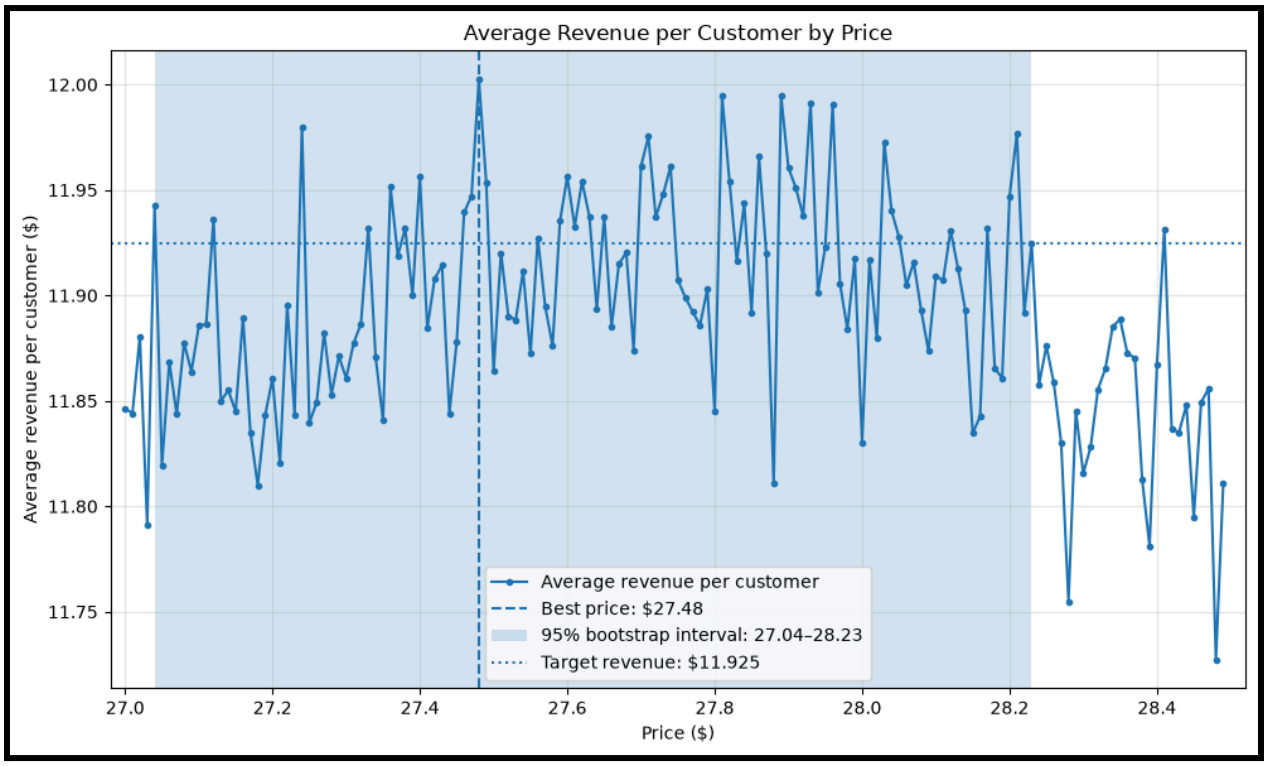
In this experiment, our goal was to find the optimal price ($0-$70) by determining which price point delivered the highest average revenue per customer in return. Since customer decisions and purchasing are random, testing each price point a single time would not accurately deliver the most optimal price. I used a simulation to evaluate prices across the range of prices between $0 and $70 by an interval of 0.5, repeating the experiment at each price and comparing average revenue per customer to determine the optimal price. After identifying a promising price region, I conducted a second search that was much more detailed, iterating across the condensed range of prices by an interval of 0.01. The benchmark for optimal expected revenue was $11.925 per customer. The overall objective of this experiment was to find the price that performs as close to this benchmark as possible while accounting for uncertainty in the simulation results.

2. Uncertainty and Selecting Prices

The first step was evaluating each price from $0 to $70 in increments of $0.5. At each price, I ran a total of 100 simulations, with the number of arriving customers drawn from a Poisson distribution with a mean of 100. Each arriving customer had the chance to purchase the product or not. For each of the simulations, I calculated revenue per customer by multiplying the price by the fraction of customers who purchased. I then averaged these revenue estimates to compare prices. Repeating the experiment reduced the influence of random variation in individual simulations and provided a broader view of how revenue changed with price. The first plot (seen below) shows that revenue was highest in the region when the price was around $27-$28. Although, it also revealed a smaller peak around $12-$13 and another local maximum at higher prices. This demonstrates why it is crucial to examine the full price range ($0-$70) rather than assume revenue would change uniformly or have only one local maximum.



After bootstrapping and constructing a 99% confidence interval, I determined that 99% of the bootstrap samples had an optimal price that fell in the range [27.0, 28.5]. So, I tested prices from $27.00 to $28.50 in increments of $0.01, running 1,000 simulations at every possible price. This gave a much finer comparison than the initial range of prices I explored. Using bootstrap sampling for each price, I repeatedly resampled its simulated revenue observations (with replacement), recalculated average revenue, and selected the price that returned the highest revenue per customer. The resulting distribution of selected prices was used to estimate another confidence interval for the true optimal price. The second plot (seen below) shows the best observed price of approximately $27.48, which had an estimated average revenue of around $12 per customer. The displayed bootstrap 99% confidence interval, using the tighter range of prices, is approximately $27.04-$28.23. This interval indicates that there is still much uncertainty around the optimal price as small differences in prices may reflect simulation noise rather than a meaningful difference in revenue, which can be seen in the plot as well.



3. Challenges Faced

One challenge I faced during this experiment was distinguishing meaningful changes in average revenue per customer from random fluctuations. Since the number of customers and their purchase decisions varied across simulations, a single experiment can produce a misleading estimate. I addressed this by repeating experiments many times at each price point and using the average revenue per customer for comparisons. I then increased the number of repetitions during the finer search to try to make estimates around the finer range more stable.

The two plots were useful for different purposes as the plot looking at the full range of prices ($0-$70) helped locate promising regions, while the detailed plot made it possible to compare nearby prices and identify the uncertainty around very small differences in price.

The second challenge I was facing throughout this process was deciding how to balance search precision, computational cost, and confidence in my result. Testing every price in very small increments across the entire range ($0-$70) would require many simulations and heavy computational costs. Instead, I decided to look through the range at an interval of $0.50 (instead of $0.01) in order to first determine the general range of the global maximum, for average revenue per customer, before looking at each individual price. After finding the global maximum,I narrowed the search to prices between $27.00 and $28.50 and used a $0.01 interval. I also increased the number of simulations from 100 to 1,000 to try to reduce the impact of random variation between small price differences. Then, I used bootstrap resampling to examine the uncertainty in the selected price. This approach helped reduce unnecessary computation while allowing me to estimate the best price more precisely.

4. Conclusion

The first step of my optimal price search tested 141 prices, with 100 simulations per price, for 14,100 total simulations. With an average of 100 arriving customers per simulation, this corresponded to around 1.41 million customers experimented with in the first search. The fine search, second step of my experiment, tested 150 distinct prices, with 1,000 simulations per price, for 150,000 total simulations, which corresponded to about 15 million customers. The two stages involved around 16.41 million customers in total. For our revenues during this experiment, the second plot, under our finer price range, reveals an estimated revenue of about $12.00 per customer at a price of $27.48. Relative to our benchmark of $11.925, this is 100.6% of the benchmark revenue. This estimate being slightly above the benchmark does not establish the true optimum as it is likely because of sampling variation. Overall, my experiment identified an optimal price near $27.50 and used a finer search and bootstrap confidence intervals to investigate uncertainty around that choice.